Proposal for Reorienting the VES Pipeline Around Ancient Greek Uncial Data

Working draft for discussion and revision. Prepared in support of internal project review and future planning.

1. Purpose of This Document

This document has been prepared by Joseph Brennan, *alias* γραμματεύς. It summarizes the present state of the VES project, an attempt at improved ink detection and character recognition begun by Adithtya Laakso, *alias* HugoHead, in pursuit of a Vesuvius Prize. It outlines reasons for concern about some of its original assumptions and proposes a set of concrete lines of inquiry for future work. It is not intended as a criticism of the original work as such. Rather, it is an attempt to clarify which parts of the system appear promising, which parts may have been shaped by assumptions that do not fit the target domain closely enough, and how the project might be redirected in a more historically and paleographically grounded way.

Many of these suggestions have been made previously -- in a less developed form -- in text and voice chats in the *Undergraves* Discord server, which served as a platform for asynchronous communication between former team members scattered across North America and Europe. This paper aims to articulate proposed changes in a more formal and detailed manner.

It is assumed that HugoHead will remain the leader of this team and will continue to participate. Other members of the team seem to have lost interest. It is hoped that if new members join the team, or old ones rejoin it, this paper will serve to explain the reasoning behind the changes that were made.

2. What Hugo Originally Created

The existing VES codebase implements a custom machine-learning pipeline aimed at ink detection on virtually unwrapped segments of ancient Greek scrolls. In broad terms, the project includes:

- a synthetic data-generation pipeline for creating papyrus-like images;
- a segmentation and classification model built around a hybrid attention-based architecture;
- a training loop designed for grayscale single-letter inputs and corresponding masks;
- visualization and inference scripts for inspecting outputs.

This work reflects real technical effort and experimentation. The current repository shows that substantial thought went into building a custom model, a synthetic-image workflow, and a training regime around it.

3. Original Intent and Current Tension

The project appears to have been built around an earlier workflow that relied heavily on synthetic or modern-handwriting-like assumptions. At some point, the effort shifted toward more realistic ancient Greek material, but the codebase still retains a number of assumptions from the earlier phase.

In addition, the ancient Greek material selected for eventual testing of output was not ideal. Even if Thucydides was intended only as a test author, I do not think he is a good default textual prior for this project. The Herculaneum villa material is overwhelmingly associated with Epicurean texts, and even if not all of the scrolls prove to be Epicurus or straightforward philosophical prose, the likely corpus still seems much closer to philosophical writing than to Thucydidean historiography.

Thucydides is a highly distinctive and somewhat idiosyncratic prose author, with a compressed, detached, and unusually marked style relative to much other Greek writing. For that reason, a language prior based on Thucydides could bias reconstruction in an unrepresentative direction, especially where the visual evidence is weak. He may still be useful as a benchmark or stress test, but not as the most plausible model for likely villa content.

The key tension is this: the current active pipeline was shaped around data and preprocessing assumptions that do not seem well aligned with the actual appearance of ancient Greek uncial letterforms as preserved on papyrus or in related manuscript imagery, or with the types of ancient Greek texts likely to be found.

4. Main Concerns

4.1 Domain Mismatch

The most important concern is domain mismatch. Ancient Greek uncials written with a reed pen or similar broad-nib instrument do not have the same stroke structure, contrast profile, or letter variation as modern Greek minuscule handwriting written with ballpoint pens. Even where labels overlap, the morphology is different.

A modern Greek handwritten text such as a letter will generally contain modern line spacing and paragraph breaks, monotonic accents, and mostly West European-derived punctuation,with French-style quotation marks (*guillemets)* and the preserved ancient interpunct, used as a semicolon. Accents are not present in the modern Greek dataset that HugoHead was using, but they are frequently present in any Modern Greek whole text samples that I have seen.

By contrast, most of the readable Herculaneum papyri use a *scripta continua* format with no accents, punctuation or spacing, but punctuation was known to the ancients, and might appear in some scrolls. Yet even in ancient texts where accents appear, they are polytonic in nature, and differ from the monotonic standard formally adopted after 1982. The same applies to aspirants and ancient forms of punctuation which may or not be present in the ancient texts, but which have been inherited by the daughter language in greatly modified form, if at all.

If the target is to help with ink detection on Vesuvius material, then training assumptions should be anchored as closely as possible to ancient letterforms and these textual realities, not merely to "Greek script" in a general sense.

4.2 Preprocessing Assumptions

Manual preprocessing experiments suggested that no single global recipe works reliably across all samples. In particular, hard thresholding and aggressive cleaning often destroy useful grayscale structure. Some samples remain readable only when tonal information is preserved.

It is not only I who have found this to be a problem. Dr. Frederica Niccolardi of the University of Naples has found that different versions of the same image provide varying degrees of readability for different text features, as outlined in her talk "Role & Challenges of Papyrological Expertise" delivered at the Michigan Institute for Data & AI in Society Vesuvius Scolls Mini-Symposium of 2025 (https://www.youtube.com/watch?v=WAWQmdBDbHg). This suggests that preprocessing should be treated as conditional and data-sensitive, not as a one-size-fits-all transformation.

4.3 Class Distribution

The extracted uncial corpus is not evenly distributed across letters. Some classes are abundant, while others are sparse. That imbalance likely reflects a combination of real language frequency, preservation bias, annotation bias, and scribal variation. For that reason, it may be a mistake to equalize the dataset itself by force. Preserving the real corpus while compensating during training is likely more defensible than flattening the data.

4.4 Geometry and Input Assumptions

The active pipeline had been operating with a fixed 128x128 image size and 32x32 mask size. The extracted ALPUB_v2 uncial images are natively 70x70 RGB images. This does not mean that training should be done at exactly 70x70, but it does suggest that the old geometry may be more of a legacy of the earlier workflow than a principled fit for the uncial source data.

4.5 GCDB Uppercase Assumptions

The training code appears to treat the GCDB-based source as an uppercase-only character set, regardless of how the Kaggle dataset may have been organized internally. In the GCDB-based generator, folder labels are normalized with **.upper(),** and the active class map is written as a single uppercase-style label set rather than as separate uppercase and lowercase classes. The original GCDB-data set from Kaggle showed lower-case variants as well as upper. The lower-case sets may have been discarded once it was realized that the target text will have only "capitals," that is, uncial letter forms, but again, Modern Greek capitals written with modern pens are no substitute for ancient uncial written with ancient pens.

5. Why the ALPUB_v2 Dataset Matters

The extracted ALPUB_v2 corpus is important because it provides a far more relevant source of ancient Greek character imagery than modern handwriting datasets. It includes:

- 205,797 total samples;
- 24 letter classes;
- native 70x70 RGB images;
- historically relevant forms such as lunate sigma;
- a naturally uneven but informative class distribution.

This corpus is therefore a better starting point for modeling assumptions, even if it introduces complications in balancing, preprocessing, and label-handling.

6. Proposed Lines of Inquiry

6.1 Re-center the Data Pipeline on Ancient Uncial Material

The first line of inquiry should be to make the uncial corpus, not the older synthetic or modern-handwriting assumptions, the primary reference point for the data pipeline.

6.2 Preserve Real Class Distributions

Rather than forcing every class to be equally large, preserve the real dataset and address imbalance during training through weighting or sampling. Preliminary analysis suggests that document-based inverse-square-root weighting may be a reasonable and non-aggressive first option.

6.3 Split by Document, Not Purely by Sample

Because many crops may come from the same document context, train/validation splits should ideally be grouped by document identifier instead of randomized across individual samples.

6.4 Treat Preprocessing as a Research Problem in Its Own Right

Rather than assuming that a single thresholding or cleaning recipe is correct, test multiple representations:

- raw grayscale derived from RGB originals;
- locally normalized grayscale;
- adaptive-threshold outputs;
- edge or contrast-derived channels;
- soft masks rather than only hard binary masks.

6.5 Compare Sample-Based and Document-Based Weighting

Initial analysis suggests that raw sample-count weighting may overcorrect, whereas document-count weighting may better reflect the true diversity of the corpus. This should be tested empirically rather than assumed.

6.6 Revisit Labels and Paleographic Variation

Some classes may contain multiple scribal variants that are visually meaningful. It may be worth examining whether the current labeling approach hides distinctions that matter for recognition and segmentation.

7. Recommended Immediate Next Steps

1. Review the active assumptions in the codebase together, especially the data and preprocessing assumptions.
2. Integrate the uncial manifest into the loader only after agreement on direction.
3. Run small, controlled experiments rather than broad redesigns.
4. Document outcomes clearly so that future changes are easy to justify or reject.

8. Concluding Position

The central claim of this proposal is modest but important: if the goal is to improve ink detection for ancient Greek material, then the project should be guided more directly by ancient Greek uncial data, paleographic judgment, and the actual behavior of real samples under preprocessing, rather than by assumptions inherited from a less domain-specific earlier workflow.

This does not mean discarding the original work. It means using that work as a base while testing whether the project's assumptions can be brought into closer alignment with the target domain.